

Comparative Study of Bioinformatics Databases (NCBI, UniProt & PDB)

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Introduction

Bioinformatics is an interdisciplinary field that combines biology, computer science, mathematics, and information technology to analyze biological data. It helps researchers understand DNA sequences, protein structures, gene functions, and evolutionary relationships using computational methods. NCBI, UniProt and PDB are among the most widely used databases for biological research.

NCBI

The National Center for Biotechnology Information (NCBI) provides free access to nucleotide sequences, genomes, genes, PubMed literature and tools such as BLAST. Advantages include comprehensive data, regular updates and integration with many research resources. A limitation is that beginners may find the interface overwhelming.

NCBI Homepage



Figure 1: NCBI Homepage

UniProt

UniProt is a comprehensive protein database containing protein sequences, functional annotations, domains and disease associations. It offers high-quality curated data but focuses mainly on proteins.

UniProt Homepage

The screenshot shows the UniProt homepage with a dark blue header. The UniProt logo is on the left, followed by navigation links: BLAST, Align, Peptide search, ID mapping, and SPARQL. On the right, it says 'Release 2026_02 | Statistics' and has icons for a storefront, a shopping cart, an envelope, and a help button. The main content area has a large 'Find your protein' heading. Below it is a search bar with 'UniProtKB' on the left and 'Advanced | List | Search' on the right. Under the search bar, it says 'Examples: Insulin, APP, Human, P05067, organism_id:9606'. At the bottom of the main area, it says 'UniProt is the world's leading high-quality, comprehensive and freely accessible resource of protein sequence and functional information. Cite UniProt'' and has a green 'Help' button. A light blue banner below the main area says 'Over 26,000 UniProtKB unreviewed/TrEMBL entries now include annotations from machine learning / artificial intelligence (AI) using a Protein Natural Language Model, ProtNLM2, developed by collaborators at Google DeepMind. This includes predictions for function comments, subcellular locations, keywords, and Gene Ontology'. At the very bottom, a black banner says 'This website requires cookies, and the limited processing of your personal data in order to function. By using the site you are agreeing to this as outlined in our Privacy Notice' and has a button that says 'I agree, dismiss this banner'.

UniProt

BLAST Align Peptide search ID mapping SPARQL

Release 2026_02 | Statistics

Find your protein

UniProtKB Advanced | List Search

Examples: Insulin, APP, Human, P05067, organism_id:9606

UniProt is the world's leading high-quality, comprehensive and freely accessible resource of protein sequence and functional information. Cite UniProt''

Help

Over 26,000 UniProtKB unreviewed/TrEMBL entries now include annotations from machine learning / artificial intelligence (AI) using a Protein Natural Language Model, ProtNLM2, developed by collaborators at Google DeepMind. This includes predictions for function comments, subcellular locations, keywords, and Gene Ontology

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Figure 2: UniProt Homepage

The Protein Data Bank stores three-dimensional structures of proteins, nucleic acids and other biomolecules. It is widely used for structural biology, visualization and drug discovery.

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256,650 Structures from the PDB archive
1,062,058 Computed Structure Models (CSM)
Enter search term(s), Ligand ID or sequence
Include CSM
Advanced Search | Chemical Search | Browse Annotations

PDB-101
OPDB
EMDataResource
NAKB
wwPDB
PDB-IHM
f x t d in

On July 21, 2021, wwPDB will fully transition to extended 12-character PDB IDs, the PDBx/mmCIF format, and the re-organized PDB archive.
Start using the PDB Beta Archive today.

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RCSB Protein Data Bank (RCSB PDB) enables breakthroughs in science and education by providing access and tools for exploration, visualization, and analysis of:
Experimentally-determined 3D structures from the Protein Data Bank (PDB) archive
Integrative 3D Structures from the PDB Archive
Computed Structure Models (CSM) from AlphaFold DB and ModelArchive

NEW Explore Integrative Structures
PDB-101 Training Resources

July Molecule of the Month
Hantavirus

Latest Entries
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Features & Highlights
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Publications

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PIRIPR bound to monoclonal antibody MAD8-2
9TUZ

Improved OneDep Support for 3D ED/MicroED
3D ED/MicroED Electron Crystallography metadata are now collected and represented using a macromolecular crystallography data model co-developed by the community.
RCSB.org Help and Documentation
Search or browse the collection of articles designed to help users explore the world of 3D biomolecular structures
Register for the July 30 Webinar on the AI-Powered 3D Structure Similarity Search
Learn how to use this new feature that delivers structural comparisons across experimentally determined and predicted Computed Structure

Meet the RCSB PDB at ISMB
Join us for posters and presentations at the Conference on Intelligent Systems for Molecular Biology. Interested in joining our team? Talk to us to learn more.
» 06/08/2026
Watch the Webinar: Exploring the Workhorses of Biotechnology
Learn how to use RCSB.org to analyze plastic-degrading enzymes
» 06/29/2026
PDB-101 Focus: Biotechnology
PDB-101 materials explore how researchers are using biology in industry. 3D print alpha-amylase, which is used in large quantities in the production of high fructose corn syrup

PDB at a Glance
256,158 Experimental PDB Structures
392 Integrative PDB Structures
82,085 Structures of Human Sequences
22,016 Nucleic Acid Containing Structures
More Statistics
992,732 AlphaFoldDB
69,326 ModelArchive

About
About Us
Citing Us
Publications
Team
Careers
Usage & Privacy
Support
Contact Us
Help
Website FAQ
Glossary
Service Status
RCSB PDB is hosted by
UC San Diego SDSC
UCSF
RCSB PDB is a member of
Worldwide PDB
EMDataResource
RCSB Partners
Nucleic Acid Knowledgebase
Core Trust Seal
Global Core BioData Resource

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Figure 3: RCSB PDB Homepage

Comparison

NCBI focuses on DNA/RNA and genomes, UniProt focuses on proteins and annotations, while PDB specializes in 3D molecular structures. All three are free resources and complement one another in bioinformatics workflows.

Applications

These databases are used in drug discovery, genomics, biotechnology, disease research, vaccine development, molecular biology and evolutionary studies.

Conclusion

NCBI, UniProt and PDB are essential bioinformatics resources. Together they provide sequence information, protein knowledge and structural data that support modern biological and medical research.

References

1. National Center for Biotechnology Information (NCBI)
2. UniProt
3. RCSB Protein Data Bank (PDB)
4. National Institutes of Health (NIH)
5. David W. Mount, Bioinformatics: Sequence and Genome Analysis